

Bangladesh Army University of Science and Technology (BAUST), Saidpur



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Project Title: FitEats

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Tanjila Sarkar

Abstract

FitEats is an Android-based mobile application developed to encourage healthy eating habits and promote nutritional awareness. The core functionality includes user registration, personalized diet plan creation based on goals like weight loss or muscle gain, a BMI calculator, nutrient recommendations, and a real-time nutrition tracker using the CalorieNinjas API. The application uses Firebase Authentication for secure login and Firebase Realtime Database for user profile management. This project aims to facilitate a user-friendly, intuitive, and educational experience, making health and fitness management accessible on mobile devices.

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Chapter: 1

Introduction

1.1 Android Development

Android is a mobile operating system developed by Google. It is based on a modified version of the Linux kernel and other open-source software and is designed primarily for touchscreen mobile devices such as smartphones and tablets. Initially developed by Android Inc., which Google bought in 2005. Android was unveiled in 2007, with the first commercial Android device launched in September 2008. The operating system has since gone through multiple major releases, with the current version being 9 "Pie", released in August 2018. The core Android source code is known as Android Open-Source Project (AOSP) and is primarily licensed under the Apache License.

Android is also associated with a suite of proprietary software developed by Google, called Google Mobile Services (GMS) that very frequently comes pre-installed in devices, which usually includes the Google Chrome web browser and Google Search and always includes core apps for services such as Gmail, as well as the application store and digital distribution platform Google Play, and associated development platform. These apps are licensed by manufacturers of Android devices certified under standards imposed by Google, but AOSP has been used as the basis of competing Android ecosystems, such as Amazon.com's Fire OS, which use their own equivalents to GMS.

Android has been the best-selling OS worldwide on smartphones since 2011 and on tablets since 2013. As of May 2017, it has over two billion monthly active users, the largest installed base of any operating system, and as of December 2018, the Google Play store features over 2.6 million apps.

Chapter: 2

About FitEats

2.1 Introduction

In the modern age of digital transformation, mobile applications have emerged as powerful tools for health management. The 'FitEats' Android application is developed to help individuals adopt healthy dietary habits by enabling them to create personalized diet plans and track their nutritional intake. The primary objective is to support users in achieving specific health goals such as weight loss, muscle gain, or maintaining a balanced diet.

2.2 Objective

The main objective of the FitEats mobile application is to assist users in managing their diet and nutrition efficiently. The goals include:

- Allow users to register and manage personal health data.
- Enable personalized diet plan creation based on goals and budget.
- Integrate BMI calculation for fitness assessment.
- Provide nutritional insights using third-party API integration.
- Ensure secure and user-friendly experience with Firebase Authentication and Realtime Database.

2.3 Literature Survey

Numerous mobile applications focus on dietary tracking and nutrition analysis. Applications like MyFitnessPal and HealthifyMe have paved the way for digital health management tools. However, these applications often involve complex interfaces or require premium subscriptions for access to essential features. FitEats aims to bridge this gap by offering a free, user-friendly, and goal-oriented platform tailored for individual dietary needs.

Chapter: 3

Software Requirements Specification

3.1 Software

3.1.1 Android Studio

Android apps can be written using Kotlin, Java, and C++ languages. The Android SDK tools compile your code along with any data and resource files into an APK, an Android package, which is an archive file with an .apk suffix. One APK file contains all the contents of an Android app and is the file that Android-powered devices use to install the app.

The Android system implements the principle of least privilege. That is, each app, by default, has access only to the components that it requires to do its work and no more. This creates a very secure environment in which an app cannot access parts of the system for which it is not given permission. However, there are ways for an app to share data with other apps and for an app to access system services:

- It's possible to arrange for two apps to share the same Linux user ID, in which case they are able to access each other's files. To conserve system resources, apps with the same user ID can also arrange to run in the same Linux process and share the same VM. The apps must also be signed with the same certificate.
- An app can request permission to access device internet and data such as the user's contacts, SMS messages, the mountable storage (SD card), camera, and Bluetooth. The user has to explicitly grant these permissions. For more information, see Working with System Permissions.

3.2 Project Components

• Activity

App components are the essential building blocks of an Android app. Each component is an entry point through which the system or a user can enter your app. Some components depend on others. An activity is the entry point for interacting with the user.

the activities work together to form a cohesive user experience in the email app, each one is independent of the others. As such, a different app can start any one of these activities if the email app allows it. For example, a camera app can start the activity in the email app that composes new mail to allow the user to share a picture.

• Layouts

Constraint Layout allows you to create large and complex layouts with a flat view hierarchy (no nested view groups). It's similar to Relative Layout in that all views are laid out according to relationships between sibling views and the parent layout, but it's more flexible than Relative Layout and easier to use with Android Studio's Layout Editor.

All the power of Constraint Layout is available directly from the Layout Editor's visual tools, because the layout API and the Layout Editor were specially built for each other. So you can build your layout with Constraint Layout entirely by drag-and-dropping instead of editing the XML.

A layout that arranges other views either horizontally in a single column or vertically in a single row. Set `android:orientation` to specify whether child views are displayed in a row or column. To control how linear layout aligns all the views it contains, set a value for `android:gravity`. For example, the snippet above sets `android:gravity` to "center". The value you set affects both horizontal and vertical alignment of all child views within the single row or column.

• Image Assets

We used Adobe XD for creating UI component backgrounds which are optimized to be used in android activities, also we used some image assets from <https://material.io/design/>

App logo is inspired from dribbble.com and redesigned in Adobe illustrator.

• Firebase

a) Firebase Real-time Database:

Firebase real-time database is used in form of collection and documents. A collection can contain number of documents and a document can contain key-value paired fields and also subcollections.

b) Firebase Authentication:

Firebase Authentication ensures secure login and account management.

Chapter: 4

Project Planning

4.1 Flow of Project

4.1.1 Dashboard activity

Dashboard activity is the launcher activity. In main activity we overridden onCreate() method which is invoked when activity is started. In the onCreate () method we check if user is register to our database by using custom flag created using shared preferences if user isn't logged in or registered then we Login activity or Register Activity otherwise we continue to dashboard activity. Dashboard activity contains Create Diet Plan, View Diet Plan, Track Diet etc. categories.

We used ui View to show navigation drawer. Clicking on any action like Home, Profile, Change Password etc. When we click on Profile button, we get image and Information of currently logged in user.

4.1.2 Project Structure:

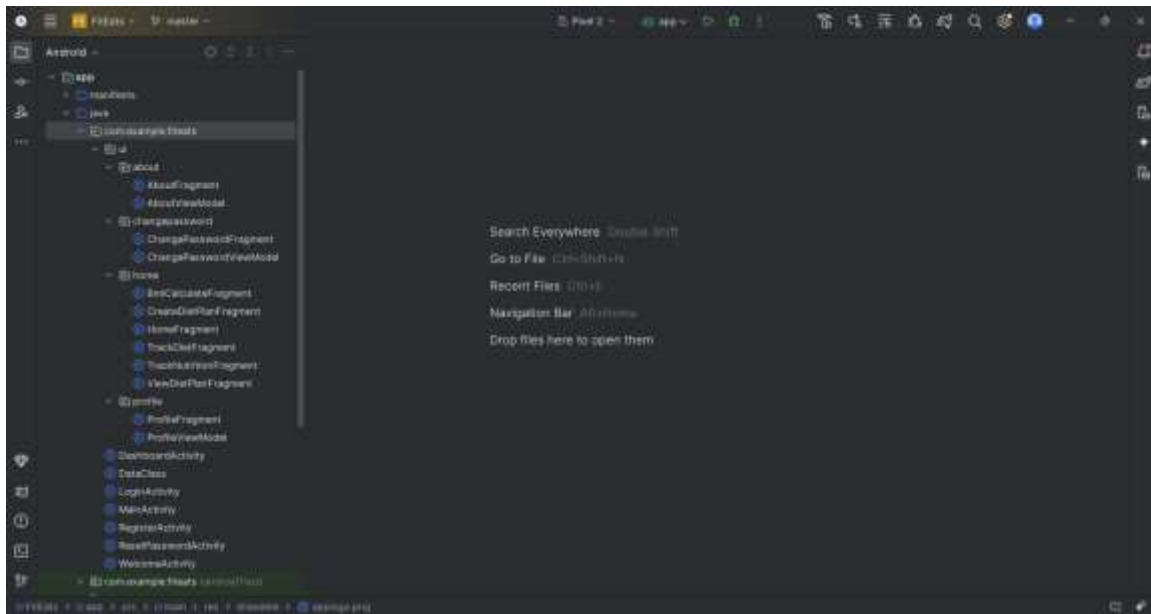


Figure: Activity and Fragment classes.

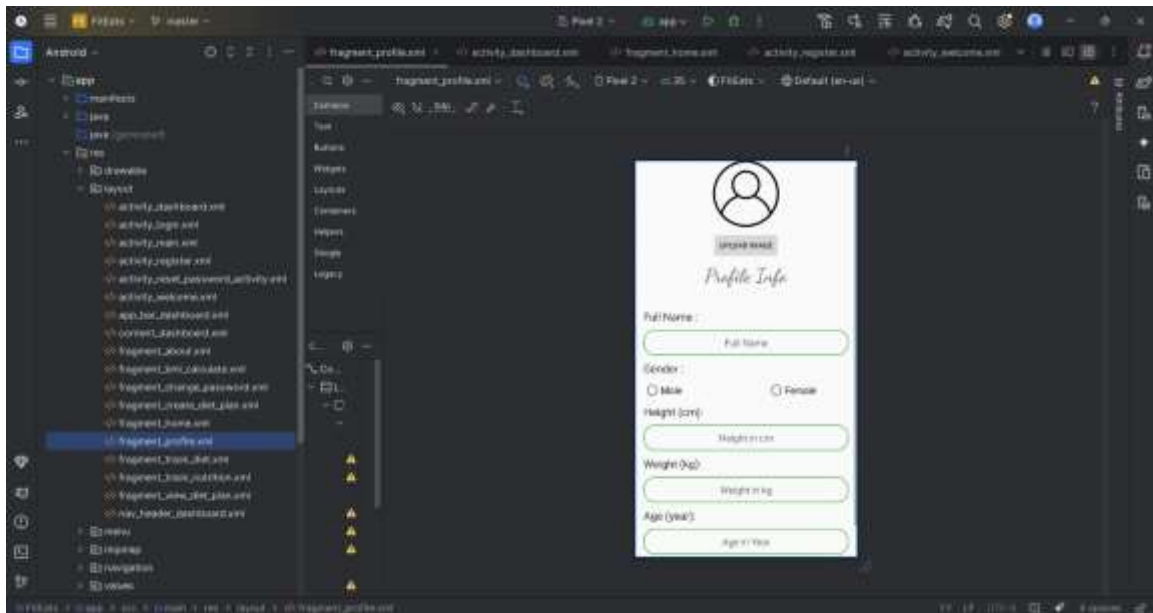


Figure: Design of Profile.

The development of the FitEats application follows the Agile methodology. It emphasizes iterative development, allowing frequent updates and user feedback integration. The core steps involve requirement analysis, UI/UX design, Firebase integration, API integration, and rigorous testing.

Chapter: 5

Design and Implementation

5.1 Implementation

The application is developed using Android Studio with Java as the primary programming language. Firebase is integrated for authentication and real-time database operations. The user interface is built using XML and Material Design components. CalorieNinjas API is used to retrieve nutritional data.

5.2 Screen Design

The application includes the following screens:

-Splash Screen



Figure: Splash Screen.

- Registration Screen

The image displays two mobile application screens for user registration. The left screen, titled "Registration", includes input fields for "First Name", "Last Name", "Gender" (with radio buttons for "Male" and "Female"), "Height in cm", "Weight in kg", "Age in Year", and an email field. The right screen continues the registration process with input fields for "Height in cm", "Weight in kg", "Age in Year", an email field, a "Enter Your Password" field, and a "Re-enter password" field. It features a green "REGISTER" button, social login options for "Facebook" and "Google", and a link for "Already have an account? Login".

Figure: Registration Screen.

- Login Screen

The image shows a mobile application login screen for "FitEats". At the top is the app's logo, which features a green apple with a muscular arm inside, and the text "FitEats" and "EAT SMART. STAY FIT". Below the logo are input fields for "Email" and "Password", followed by a green "LOGIN" button. At the bottom, there are links for "Forgot Password" and "Don't have an account? Register".

Figure: Log in Screen.

- Dashboard with Navigation Drawer

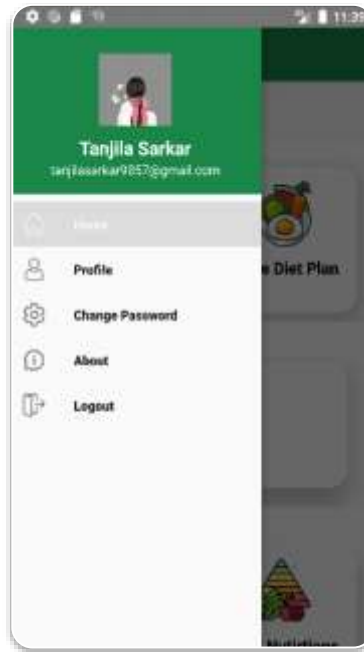
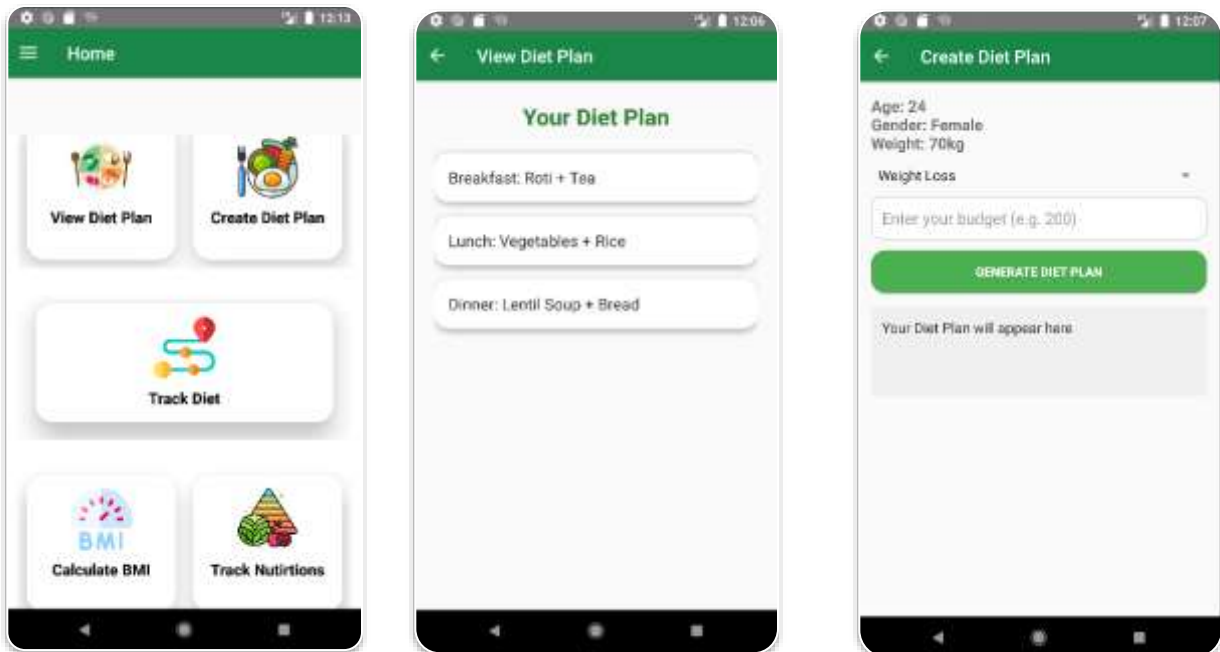


Figure: Navigation Drawer Screen.

- Home Screen with options for Diet Plan, Nutrition Info, BMI Calculator, and Tracking



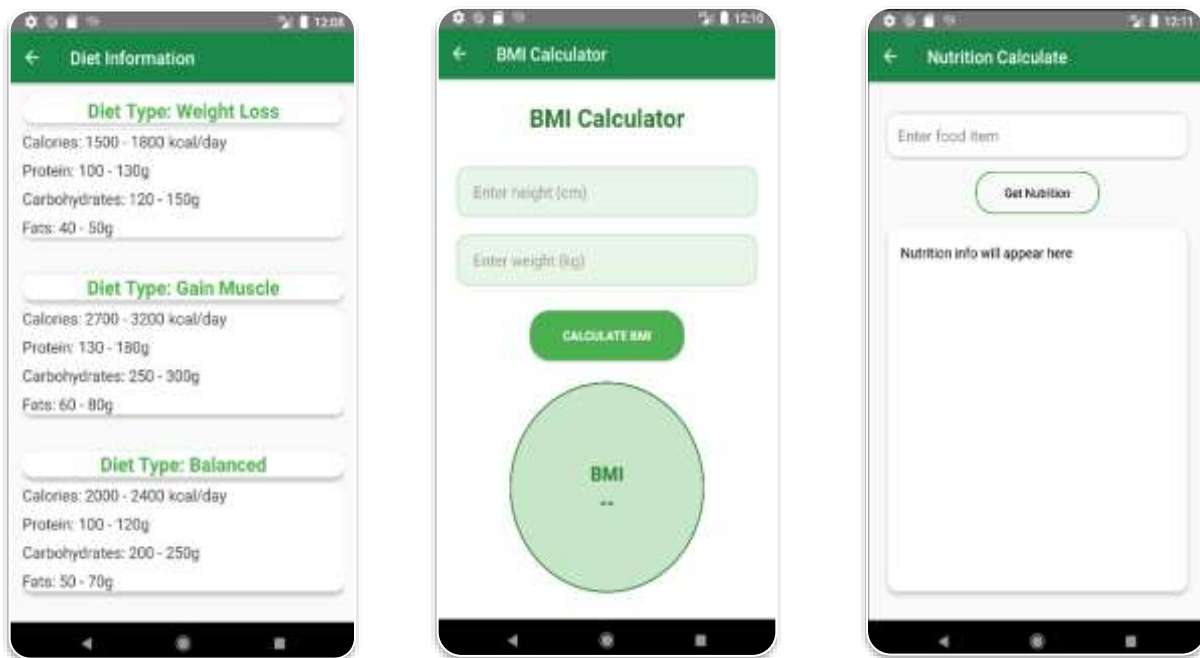


Figure: Home Screen.

- Profile:

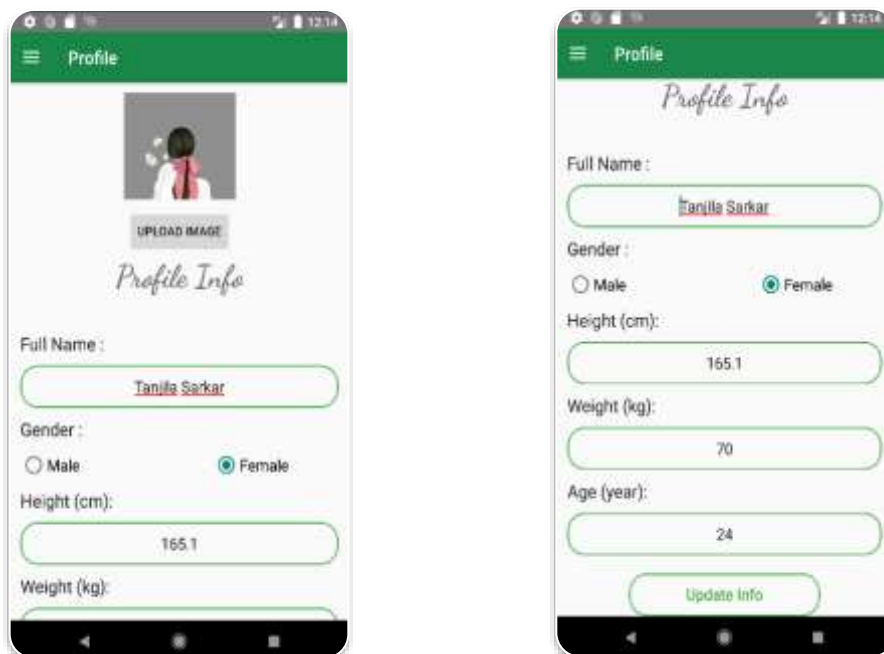


Figure: Profile Screen.

-Change Password:



Figure: Change Password Screen.

-Forget Password:



Figure: Forget Password Screen.

Each screen is developed using XML with responsive and user-friendly design principles.

5.3 Firebase Integration

To ensure a seamless and secure user experience, FitEats is integrated with Firebase, one of the most powerful mobile backend platforms. Firebase Authentication handles user login and registration securely, allowing each user to create a personalized account using email and password. Meanwhile, all essential user data such as profile details, personalized diet plans, and nutrition tracking information are stored and managed using Firebase Realtime Database. This enables instant data synchronization, so any changes made by the user are reflected in real time without the need for manual refresh or reload.

By using Firebase, FitEats achieves speed, reliability, and scalability, making it easier to manage user data, support cloud connectivity, and deliver a smooth mobile-first experience. It forms the core of how data flows through the app—efficiently and securely.

-Firebase Integration

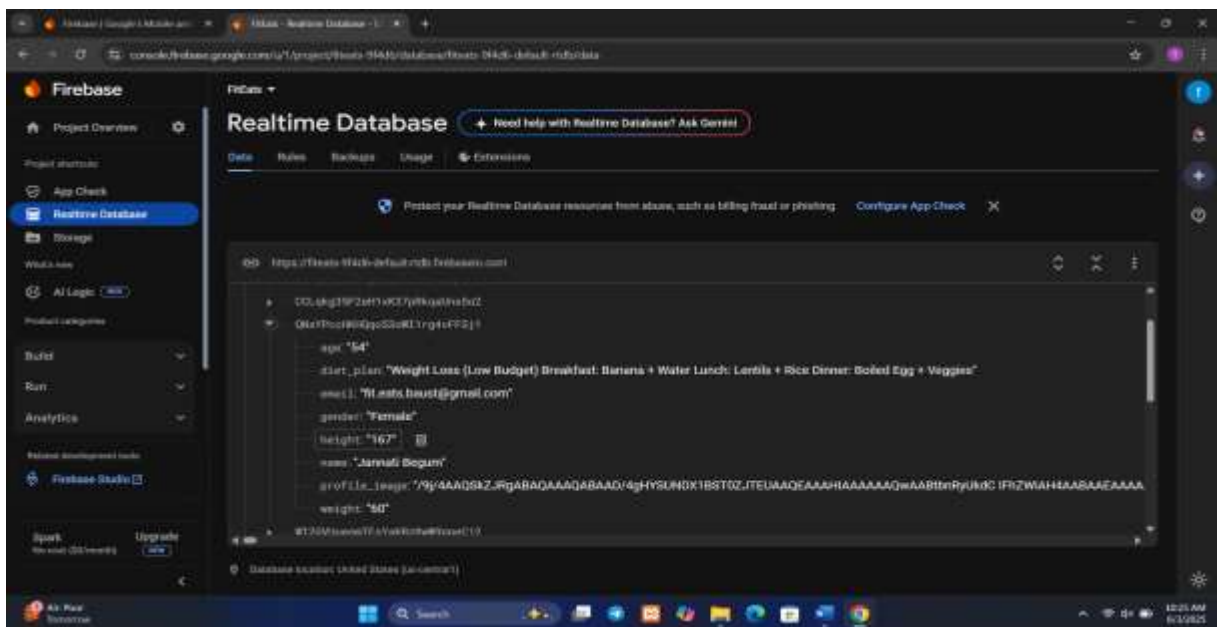


Figure: Firebase Integration.

5.3 Technologies Used

- Programming Language: Java
- IDE: Android Studio
- Database: Firebase Realtime Database
- Authentication: Firebase Authentication
- Third-Party API: CalorieNinjas API
- Design: XML, Material UI

Chapter: 6

System Evaluation

6.1 Testing

Testing was conducted at multiple levels:

- Unit Testing: Verifying each module such as BMI Calculator and Diet Plan creation.
- Integration Testing: Ensuring proper interaction between Firebase and the UI components.
- UI Testing: Validating user navigation and layout rendering.
- Performance Testing: Ensuring smooth data retrieval and real-time operations.

6.2 Security and Validation

- Email Verification
- Password Validation
- Secure Login using Firebase Authentication
- Forgot Password Email-based Recovery
- Real-time secure access to user data

6.3 Features of Application

- User Registration and Login
- Persistent Login using Shared Preferences
- Personalized Diet Plan Creation
- Budget-based Food Suggestions
- BMI Calculation Tool
- Nutrition Info via CalorieNinjas API
- Profile Management
- Firebase-based Authentication and Data Storage

6.4 Limitations

- Limited Offline Access
- Dependency on Third-Party API for Nutrition Info
- No Machine Learning based personalized insights

Chapter: 7

Conclusion and Future Enhancement

7.1 Future Enhancement

- Integration of AI for meal suggestion
- Offline data caching
- Integration with fitness wearables
- Social features for user interaction

7.2 Conclusion

FitEats is a comprehensive and user-friendly mobile application aimed at promoting healthier eating habits. It simplifies diet planning through tailored suggestions and nutritional tracking. The app combines Firebase services, external APIs, and Android technologies to provide a reliable health management platform.

References

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